

A thinking toolbox for data science practitioners

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As a data scientist, there are lots of skills you must master and different tasks you need to tackle. Many of them are technical and depend on your knowledge and experience generating data, cleaning them up, analyzing them, applying models and so on.

However, some key tasks demand one additional skill set.

You will often have to come up with new ideas and proposals that may add value to your firm or enhance your portfolio. You already have the experience of deciding on an initial portfolio project and such a process will be an important part of your career going forward. You may think that your job will entail being given a precise task, a data set and then an expected result. That may happen from time to time but most data science positions will expect you to be proactive and propose high-value projects, even as a junior data scientist. And of course, you will always take part in finding out and assessing solutions to proposed problems.

You will also have to pitch your ideas, that is, present them and defend them in front of your team, customers or other firm departments. You will also have to offer constructive criticism to projects presented by coworkers.

Besides, you will have to lay out the conclusions you reached through your work. It is not enough to just present the raw data and results (*evidence*, as Williams et al., 2003, call them) but also *reasons*, i.e., generalizations and extrapolations of those results. Once more, you will be expected to defend your conclusions and assess those from your coworkers.

All those endeavors benefit from acquiring what we call *thinking* skills or tools. All of you can think, so it is not about teaching you to do so, but providing some heuristics that may help you get out of blocks, avoid some common errors and master some techniques to give structure to your thinking process.

Structure of the workshop - I

The best way to learn the tools is by applying them. The concepts that we will be learning are not the main point: they act as a way to point your attention to the parts of the thinking process that you may find sometimes hard, such as what to do when you can't come up with a good idea or how to pinpoint something that's missing or doesn't make sense in a line of thought. Use any idea or technique as you find them useful; they are tools in your toolbox.

Think of the workshop as a hands-on tutorial, say about the Pandas library, where we discuss a couple of basic concepts and then go on to work with real data.

Along the workshop we will be finding ideas for projects, preparing pitches and discussing results.

However, in this document I will present the main concepts for reference, combining some theory (not a lot, promised!) and practical guidelines.

A little bit of theory: Reasoning is a conversation

One part of our thinking toolbox falls under the category of *reasoning*. It is typically **not** the first tool that is required in our *thinking pipeline*, that would be a tool for generating ideas; reasoning is more about defending and assessing them. However, we will begin here because it sets the tone for how we tackle thinking techniques in general.

Reasoning is all about convincing people; either yourself, your team or third parties. It is based on *arguments*, not in the sense of verbal fighting, but as structured sets of claims and reasons for them (among other elements).

Traditionally, reasoning has been the subject of logic and logic is mostly presented as *formal logic*. However, real tasks are better handled through more down to earth, applied *informal logic*.

The goal here, as famous philosopher Stephen Toulmin (2003a) put it, is not to be *rational*, that is, to come up with flawless, universal arguments, but to be *reasonable*: to find practical solutions suitable for the task at hand.

Informal logic can be viewed as a problem solving process (Williams et al. 2003) and we can think of how it unfolds as a conversation.

It was also Toulmin (2003b) one of the first to propose a model of reasoning along those lines. To understand it better I will present a toy example, based on a very old idea, which will help us understand the difference between formal and informal models.

One of the oldest formal models of logic is the syllogism, proposed by Aristotle. A typical example of it is:

1. **All human beings are mortal**
2. **Socrates is a human being**
3. **Therefore, Socrates is mortal**

Syllogism theory considers itself a form of *deduction*, that is, it proceeds from general ideas to particular ones, which theoretically would allow it to come up with sound, universal arguments.

According to the theory, sentence 1 would be a major premise; sentence 2, a minor premise; and sentence 3, the conclusion. Together they would make up a valid syllogism and, thus, a flawless, rational way of reasoning.

Toulmin criticizes that idea: he would say that, in practice, arguments do not unfold like that. Toulmin mentions this example and the shortcomings of classical formal logic at analyzing it but does not use it to illustrate his own model. Let's do so here.

Often the first thing that comes up in conversation is the conclusion, which Toulmin would call **claim**. Someone may say:

- a) **Hey, you know that man, Socrates. Someday he'll die.**

If the listeners agree, they'll just nod and the conversation will proceed to other topics.

Technically, that would not even be an argument. In a toy example such as this, you can be almost sure that this is what would happen.

However, let's suppose that is not so. For some reason, someone is suspicious and asks:

- b) **How do you know that? What makes you think so?**

Toulmin would call that a *rebuttal*, but we will follow a later, more detailed model by Williams et al. (2003) and call it an **acknowledgment and response**.

An unwritten rule of conversation is that the original speaker (or poster, or writer or content creator) should then state why they made that claim, maybe:

c) Well, because Socrates is a human being, you know.

Toulmin would call that *grounds* or even *data* or *evidence*, but Williams et al. would rather consider it a **reason** and reserve the name *evidence* for something else that we will discuss shortly.

Once more, if everybody agrees, the conversation is over and it would be ridiculous for normal people to go on from here. However, let's be ridiculous for a little bit and consider that someone is still not convinced. There are several variations for a new acknowledgment and response. For example:

d) What relationship is there between Socrates being a human being and him dying?

The answer could be:

e) All human beings die

That would be a **warrant**. Warrants are links between claims and reasons. They typically are not stated explicitly because they are taken for granted but they are always present in the background and it may be crucial to think about them, particularly when our audience does not share our knowledge or cultural background.

There are other possible *acknowledgements and responses* (all similarly absurd for this example, but bare with me!):

f) How do you know that all human beings die?

Reply:

g) Every single human being in recorded history has died.

Toulmin would call sentence g a *backing*, but Williams et al. point out, quite accurately, that this is simply a **reason for the warrant**: it is as if the warrant "all human beings die" had become a new claim and we ask for reasons for it. In general, we very often need such a process of **stacking simple arguments**, asking for and providing reasons for warrants or reasons for reasons, or reasons for reasons of reasons, ... you get the idea. You may also need and supply warrants for warrants or reasons. For example someone may ask:

h) The fact that all past people died really implies that all present or future people will die? [which is a valid point]

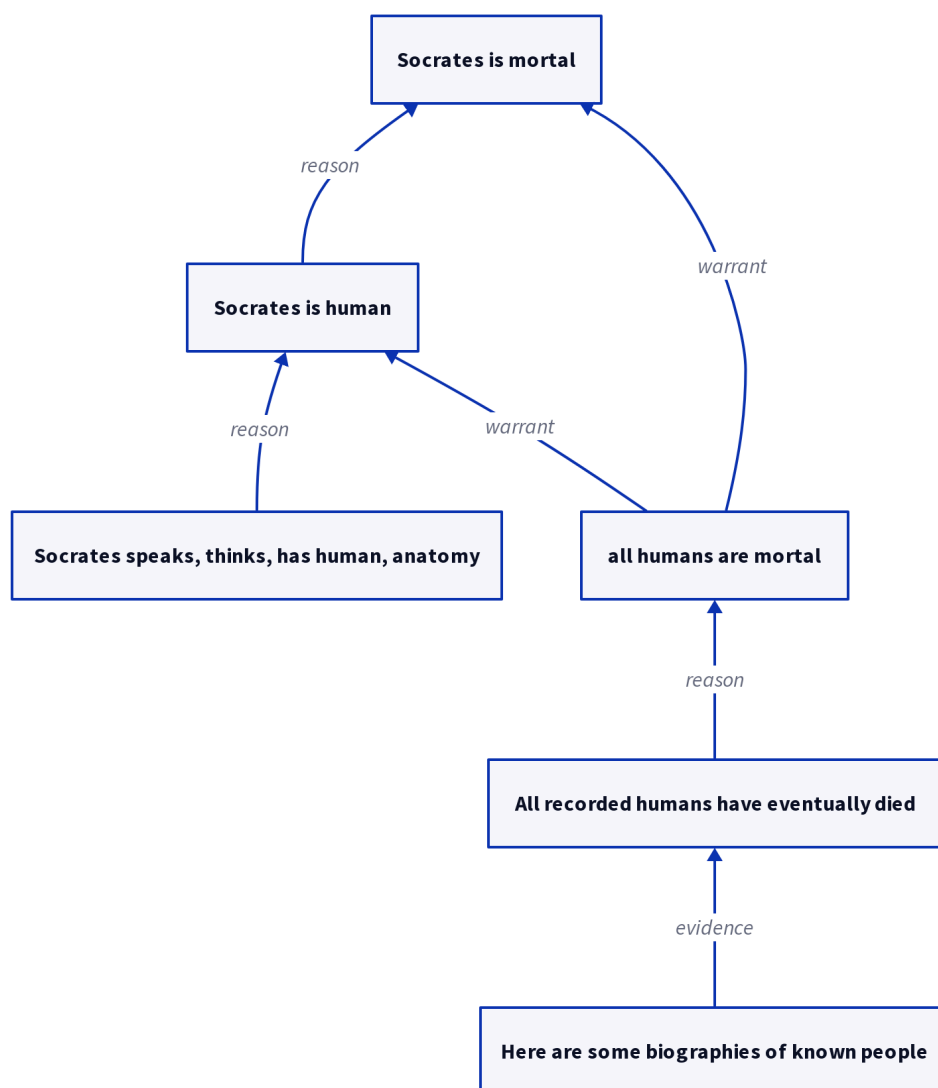
If someone would doubt, on the other hand, our reason for the warrant, i.e. sentence g (which, in fact, is what would make more sense in this discussion):

i) Are you sure that every single human has died?

It is not a reason which would make sense as a reply, but **evidence**; where reasons are more general, evidence is a concrete piece of data:

- j) **Here you have biographies of former well-known people: all of them ended up dying**
- or
- k) **Here are stats from past centuries, notice the death count.**

This is a diagram of parts of the example argument (made with [D2](#)). Notice that the warrant joins a claim and a reason.



One important distinction is between *actual evidence* (the deaths) and *reported evidence*, how actual evidence is presented, as quoted biographies or estimated death stats: it is quite possible that the report misrepresents actual evidence. This point is particularly important for

data science. In data science, evidence is typically reported as summary statistics (mean, standard deviation, explained variance, accuracy, ...) or as graphs. Actual evidence is real patterns in the data.

The conversation can go on indefinitely, combining claims, reasons, warrants, reasons for reasons, evidence for reasons, reasons for warrants and so on, in response to objections (acknowledgement and responses).

The end comes when those that take part in the conversation reach common ground and, thus, an agreement and a conclusion or, failing that, when they *agree to disagree*.

Arguments, in this sense, are collaborative and examples of *civil conversation* (Williams et al., 2003). The aim is to come up with useful, reliable conclusions, not to score points or be right at the expense of the other part. We do not insist on arriving at any final, absolute truth, which is seldom possible.

Because of the realistic, *reasonable* goal of reaching agreements and practical solutions, it is possible to argue not only about facts (as in the example) but also about goals, ethical issues (which are often important in data science) or even aesthetic judgements: it is all about presenting relevant reasons and warrants for our claims.

Reasoning in practice

So, reasoning is a conversation. How does such a conversation take place in real settings? Sometimes you do have a conversation as outlined in the last section; particularly when you are defending your proposals and your audience puts questions about it.

When you are preparing your project, or pitch, or your presentation of data, you have to carry the conversation with yourself or within your team, if you are collaborating. You will have to anticipate as many objections (acknowledgement/responses) as possible and find ways to counter them.

Such a conversation is called a **deliberation**. This is the main method we will be using to tackle tasks. Notice that a deliberation is different from a *debate*. Debates have two sides and there is an implicit rivalry between two points of view (*is this good, is this bad?*). In contrast, deliberations tackle open-ended questions (*How can we solve this? What should I do? ...*) and their goal is to find agreements and solutions. There may be specific debates in the course of a deliberation, but they are secondary to the main goal of finding a consensus. To be more specific, a typical task begins with a problem, a goal to achieve, a decision to take. As explained before, we often are asked to propose the task or problem and we will talk about how to proceed in that case later on. For now, let us assume that the problem is given.

Alec Fisher (2004) proposes a way to proceed: the **assertibility question**.

In its general form, it refers to possible reasons or evidence for our claims. It can be stated as:

What reasons and or evidence would justify this claim? (form a)

For the case of a problem, it can be changed to:

What course of action would solve this problem? What would be an acceptable solution? Why? (form b)

During our deliberation we should go back to either form of the assertibility question and then use the answers to it as a guide to look for further solutions, reasons or evidence. It is important to answer the assertibility question (form b), at least partially, before proceeding with the deliberation, so as to avoid getting misled, besides the utility of having a starting point for the deliberation. One way to understand the risk of proceeding without considering

the question is to compare the situation with a different kind of argument: an apology: if someone makes a mistake or plainly does something wrong, and then apologizes, it helps to decide beforehand what an acceptable excuse would look like, what points it should address. Otherwise, it is easy to fall for pseudo-apologies (non-denial denials, appeals to emotion, victimism, and so on).

An example

Let us suppose your firm wants to improve its brand reputation.

You ask the assertibility question (form b):

What would be an acceptable solution?

Let's suppose this is a complicated goal and you come up with a partial solution:

It would help to first assess what the brand reputation is now and why.

This may be perfectly acceptable: you are setting a sub-goal, but don't forget to revisit the question later on.

Now you may state an acknowledgement/response:

How can that help? Is it really a good idea?

Now you must find reasons in favor (or against) the idea. As a data scientist, you do not just think hard about the problem, you must do research to find reasons and evidence. You may review the state of the art for that kind of problem and the challenges you may face.

Say you read about *sentiment analysis* and when it may work and when it can fail. These are interesting reasons to take note of.

Let us further suppose that the success of the technique depends on the type and quantity of the available data. This sets up a new claim:

We can gather quality data for the task.

Then you carry on the research, this time finding out what data is available, whether you can gather some new one and performing *exploratory data analysis* on the data you have available. The results of your research are reasons or evidence that you can use in a new reasoning cycle.

Notice that you don't have to stick to this process all the time: sometimes you'll find that the deliberation proceeds smoothly and there are many good ideas, you can just go with the flow then, but do refer back to your current assertibility question to make sure you are making progress.

Sometimes, however, you may get stuck, that is the moment to go back to the reasoning process, either restating an assertibility question or finding new reasons, evidence or warrants.

Along the way you do research either in the data science literature or by analyzing your data to answer the questions that may come up.

If you record each part of the process, at the end you will have a comprehensive, well motivated proposal for action that you can present and pitch.

You can apply a similar process to other phases of your data science tasks: as you begin developing your solution, when you prepare a report about the results you found, ...

Of course, you will have to alternate this reasoning process with other elements of a proper data science solution: proper testing, good coding practices, sound choice of models, data visualization and so on.

Coming up with ideas: history and research

As we have mentioned several times, reasoning is a great tool to assess and elaborate ideas but it does not address how to come up with those ideas in the first place. Sometimes ideas come easily, through sheer inspiration or by means of our knowledge in the domain at hand. Quite often, however, we feel stuck.

Is there a way to help us generate ideas? A thinking tool that could assist us? Of course, there is no bullet proof method but there are heuristics that can help you overcome blocks. I am going to present one of them that I have found useful in many contexts. The detailed method is my own proposal but it is based on previous proposals and research, just as we used work about informal logic for our discussion about reasoning. I think it is useful to briefly review such references.

George Pólya was [an important mathematician](#), who wrote [a very influential book](#) on mathematical education (Polya & Pólya, 2014). His goal was to teach how to think mathematically and solve problems, rather than just learn formulae and procedures. He proposed a system divided into several parts and provided lots of interesting example problems in his book, whose solution you can find yourself with the help of Pólya's guidance. A key part of Pólya's proposal was a set of *heuristics*, i.e., general problem solving principles that are introduced to the student by means of a series of questions. In the beginning, the teacher would pose those questions to guide the student but overtime the student would internalize the questions and, thus, the principles. The questions do not refer to any specific problem but may apply to a large set of them. For example:

- *Can you find a problem analogous to your problem and solve that?* [analogy heuristic]
- *Can you vary or change your problem to create a new problem (or set of problems) whose solution(s) will help you solve your original problem?* [variation heuristic]

You can find more examples at the [Wikipedia article](#) on the method.

Pólya's method works very well for problems that have clearly-defined solutions, such as in mathematics, computer science or physics. However, very often we need more open problems, to which there are no unique solutions but rather a set of decisions with varying levels of potential success. Finding an interesting project or selecting a data science model belong in such a category.

To breach the gap between that kind of problem and Pólya's methods, we will use a body of research by late Ronald A. Finke (e.g. 1989) from Texas State University. His initial research was about visual imagery. Although he later extended it to other domains, we will present those first studies as a motivation.

He presented his subjects with a set of simple forms, which acted as *visual primitives*. In one version, the primitives are simple 2-D shapes, letters, numbers and symbols. In a typical task subjects were shown the primitives. Then a set of them was selected at random and communicated to the subjects, who were to mentally combine them into *interesting and potentially useful* images, by rotating, scaling and joining the primitives, without altering their shape. Subjects would draw the resulting image, called a *preinventive form*. Finke et al. (1988) presents a different set of studies but it shows examples of one version of 2-D primitives, and labeled images that combine them, and it is available online (link in references).

After coming up with preinventive forms, subjects were given a random category (e.g. *furniture, weapons, vehicles, presents, appliances, ...*) and asked to come up with a function for their form within that category.

Several independent judges rated the resulting images for interest and originality. Inter-judge reliability was measured and different variations of the task were compared using a chi-square statistical test.

Some relevant results of Finke's research were:

- Better results are obtained when the primitives are selected randomly than when subjects were allowed to choose them.
- The same is true for the categories: it's better to select them at random than allow a free choice.
- When the images of the primitives were withdrawn before communicating the chosen set, results were better than when the primitives were present during the task.
- It is better to come up with the preinventive form before knowing what category would be selected than to know the category beforehand.
- The best categories were neither too general nor too specific.
- Subjects don't get better at the task over time. There are just conditions that make it easier to come up with interesting ideas.

We can summarize Finke's results as follows: the best way to come up with interesting ideas is by alternating phases of freedom and phases of constraint.

Finding ideas in practice: beyond brainstorming

You may have heard about the technique of *brainstorming*: potential ideas are proposed spontaneously without criticizing them before beginning to discuss their suitability for the task at hand.

Brainstorming has some advantages, such as providing a nonjudgmental environment and a way to generate a number of possible ideas. However, it is often insufficient for finding high-quality ideas, particularly in a demanding, technical domain such as data science.

I am going to propose a different approach that provides a similarly nonjudgmental environment but which provides much more structure and flexibility and allows the inclusion of domain-specific knowledge. It is inspired by Polya's heuristics and Finke's research but the specific method and its details are my own proposal. I have used it in my work in different areas and I have found it useful.

The idea is to come up with several *layers*, each composed of different types of *primitives* that are relevant for the domain that we are working on and ask questions and/or conduct a bit of research about each of them. In Finke's research there are two layers: visual primitives and categories, each one with different choices: either visual shapes or the different categories (furniture, ...). You can see examples in the link provided below in the references section (Finke & Slayton, 1988).

Notice that the two layers differ not only in the choices but also in their overall *modality*: either visual or function.

At the end of each stage of the procedure we commit to one choice (or sometimes a couple of them) from each layer either at random or because it seems appealing, but without criticizing it too much. This procedure is about exploring new possibilities

The procedure works best if we find a set of layers appropriate for the domain in which we are working. It is better to use several very distinct layers (certainly more than two). The mere fact of deciding the layers and primitives is a very helpful way of developing ideas. It is also advisable to make them as specific as possible.

That leaves a lot of possibilities, which depend on the current job and kind of problem.

For a general data science project, such as for a portfolio, one possibility may be:

- **First layer:** find problems. Possible questions/pointers: What do you think it's not working? In the world or in technology. What concerns you? Conduct a search for issues people complain about (e.g. in Reddit), either globally or locally.
- **Second layer:** Go deeper. What's at the base of the problem? Go down the rabbit hole. Difficulties with self-driving vehicles may point to a deeper problem about how to organize transport; issues about reading applications may be related to how we can reconcile monetizing authors' work and easy access to information, ...
- **Third layer:** search for past solutions/projects that might be related to the problem. For example research portfolio projects from past batches. This would provide you with a lot of ideas for your project.
- **Fourth layer:** potential datasets. Find several available datasets that might be useful for solving your problem. Discuss how they can be improved and whether you should collect new data.
- **Fifth layer:** data science models that may be relevant for the chosen dataset and problem: reinforcement learning, large language models, convolutional networks, ... Once more, this implies a bit of research.
- **Sixth layer:** what data product can we design from the results of our model?

Typically, you would repeat the process a couple times to compare alternatives.

At the end of the process, you have a few candidate ideas. Commit to them for a bit without judgment. Maybe write a small *elevator* pitch for it.

But don't forget they are just drafts. It's not about finding a good idea, but generating a bunch of them. This is where the *brainstorming* part of the process kicks in. Don't censor yourself at this stage. Once you finish considering the ideas, you will have to take another hard look at their rough edges and the cons. Consider how you can improve on them. You may have to go back to the choices you discarded previously and change the project substantially. You may have to repeat the process and find out

Once you finish, you are free again to make any changes you want, including going back and picking (or combining) some of the discarded choices.

The goal is to provide some interesting, thought provoking and well motivated candidate projects for you to choose.

Structure of the workshop - II

So, what will we be doing in the workshop? It will have two parts. On the first day, we will be generating candidate ideas. The next day we will discuss them; discard many, change and a few and assess how much value we can get out of them.

We will be using the techniques discussed in this document plus other data science skills throughout the workshop.

Outro

I hope this document and the workshop are useful. I am looking forward to meeting you in person.

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